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Online Data-Driven Evaluation Method of Static Voltage Stability Margin for AC/DC hybrid Power Systems Considering Uncertainty of Renewable Energy

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***Abstract*—The increasing integration of renewable energy and HVDC technology has significantly heightened the operational complexity of large-scale power systems and amplified the risk of voltage collapse. To enable rapid and effective voltage stability assessment of such highly uncertain AC/DC hybrid systems, a data-driven online probabilistic predictive framework for the static voltage stability margin (SVSM) is proposed. A LightGBM–SHAP–KDE algorithm is developed within this framework, where SHAP is employed for feature selection and explanatory analysis, LightGBM is utilized to ensure prediction efficiency, and KDE extends point predictions to probabilistic distributions, thereby enabling accurate, fast, and interpretable probabilistic SVSM prediction. Case studies on the modified IEEE 39-bus system and the Polish 2383-bus winter-peak system demonstrate that the proposed approach achieves superior accuracy and computational efficiency compared with EXtreme Gradient Boosting (XGBoost), random forest (RF), and support vector machine (SVM), and also achieves better real-time performance compared with the traditional Cumulant-based interval calculation method. Additional evaluations under PMU/WAMS data latency and asynchrony, topology changes, measurement noise and missing data further confirm the robustness and scalability of the proposed framework under complex operating conditions.**

***Index Terms*— static voltage stability margin, renewable energy, probabilistic prediction, LightGBM-SHAP-KDE, AC/DC hybrid power systems.**

NOMENCLATURE

Abbreviations

SVSM	Static voltage stability margin
LightGBM	Light Gradient Boosting Machine
SHAP	SHapley Additive exPlanation
KDE	Kernel Density Estimation
PMU	Real-time phasor measurement unit
VSC	Voltage Source Converter
PV	Photovoltaic
RF	Random Forest
SVM	Support Vector Machine
WAMS	Wide area measurement system
XGBoost	EXtreme Gradient Boosting
HVDC	High voltage direct current
CPF	Classic methods include continuous power flow

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I. INTRODUCTION

WITH the large-scale integration of renewable energy sources such as wind and photovoltaic generation, together with the extensive deployment of high-voltage direct current (HVDC) technology, modern power systems are undergoing profound structural and operational transformations. While these developments promote sustainability and energy diversification, they also introduce new challenges to ensuring secure and stable grid operation. Among these challenges, static voltage stability has become particularly critical, as the risk of voltage collapse increases significantly in complex AC/DC hybrid systems with high renewable penetration. Addressing how to maintain voltage stability under such conditions has become a key issue in both power system research and practical operation [1].

The Static Voltage Stability Margin (SVSM) is widely used as an indicator to assess the degree of static voltage stability, representing the maximum load increase that a system can withstand under given operating conditions. With the large-scale integration of renewable energy and the rapid development of AC/DC hybrid architectures, SVSM estimation faces new challenges, including high uncertainty, interdependent constraints, and stringent real-time requirements. In recent years, research on SVSM evaluation has mainly advanced along the following directions: traditional model-based computation methods, deterministic data-driven prediction methods, and probabilistic prediction methods.

Traditional model-based computation methods of SVSM include Continuous Power Flow (CPF) [2], Thevenin Equivalent Method [3], Sensitivity Analysis Method [4], and Modal Analysis Method [5]. Although these methods have high computational accuracy and strong interpretability, they are complex to model and have high computational costs, making them unsuitable for real-time monitoring and early warning of power grid operation status. To alleviate these drawbacks, several acceleration strategies have been reported, such as the enhanced CPF of Kim et al. [6] and the energy-function sensitivity hybrid method of Zhang et al. [7], have accelerated calculations. Despite these advances, physics-based methods remain challenged by stochastic variability and frequent topological changes in large-scale systems, and their scalability

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to high-dimensional data and online applications.

With the increasing deployment of wide-area measurement systems (WAMS) and phasor measurement units (PMUs), data-driven approaches have been widely adopted for online evaluation of the SVSM. Typical methods include convolutional neural networks (CNNs) [8], extreme learning machines (ELMs) [9], and back-propagation neural networks (BPNNs) [10]. Akbari et al. proposed an optimized adaptive neuro-fuzzy inference system (ANFIS) that reduced the mean square error by 23% on the IEEE 118-bus system [11]. Liu et al. developed a correlation-detection random-bit forest (CD-RBF) framework capable of margin prediction within 0.1 s, though its interpretability was limited [12]. Cui et al. integrated multi-CNN architectures with a variational Bayesian (VB) inference scheme to enhance voltage stability assessment [13]. Compared with traditional computation methods, these data-driven prediction methods provide faster computation and stronger generalization, but they still struggle with inadequate representation of renewable energy uncertainty and limited interpretability.

In recent years, various approaches have been proposed to address the uncertainty in SVSM prediction arising from renewable generation variability. Representative techniques include Monte Carlo simulation (MCS) [14], probabilistic analysis [15], and interval analysis [16]. Q. Wang et al. [17] applied a probability-box model combined with interval-based Gram–Charlier expansion to characterize the fluctuation range of SVSM under variable wind and solar conditions. An enhanced ELM–KDE model was proposed in [18], where the distribution of power prediction errors was used to generate probabilistic SVSM estimates at different renewable penetration levels. T. Liu et al. [19] developed an extreme learning machine incorporating a Box–Cox transformation for SVSM interval prediction. While these evaluation methods considering uncertainty are more practical than deterministic data-driven prediction methods, they still suffer from limited interpretability and have paid little attention to DC network constraints.

As modern power systems evolve toward AC/DC hybrid configurations, SVSM prediction under high renewable penetration must not only capture wind–solar uncertainty but also incorporate VSC-HVDC control modes and DC network constraints. This markedly increases modeling and computational complexity. Ma et al. [20] developed a probabilistic continuation power flow model for assessing SVSM distribution amid wind and load variation. Liu et al. [21] proposed a bi-level OPF (BLOPF) framework to estimate worst-case SVSM within converter limits, though with limited real-time capability. To improve speed, Xia et al. [22] proposed a PPF-DKDE scheme for rapid sampling and probabilistic density estimation in hybrid grids, while Tu et al. [23] introduced a three-stage optimization using Galerkin parameterization for interval-based SVSM evaluation. Zhang et al. [24] proposed a CNN–LSTM–based framework for SVSM prediction in AC/DC hybrid systems with high renewable penetration, which demonstrated high accuracy and strong scalability for application to large-scale systems. Poursaeed et al. [25] developed a CNN–BDLSTM–based framework for voltage stability assessment in inverter-based resource (IBR)-dominated AC/DC systems, explicitly capturing the impacts of power electronic dynamics under renewable variability.

Overall, despite significant progress in evaluation methods considering uncertainty and AC/DC hybrid systems, important challenges remain. Physics-based approaches are often computationally intensive and difficult to scale, whereas data-driven models frequently sacrifice interpretability. These limitations highlight the necessity of developing SVSM prediction methods that are interpretable, probabilistic, and suitable for real-time application—providing the motivation for the methodology proposed in this paper.

This paper proposes an online probabilistic SVSM prediction method for AC/DC hybrid power systems considering renewable energy uncertainty, namely the LightGBM–SHAP–KDE method. The approach leverages real-time operational data from power grids as model inputs and provides high-accuracy, fast, and interpretable probabilistic estimates of SVSM. To further highlight the contributions and innovations of this work, Table I summarizes the comparison between the proposed method and representative approaches reported in the literature.

Different from [2] – [25] in Table I, the proposed method not only achieves high accuracy and real-time performance in SVSM prediction, but also offers strong interpretability. In particular, SHAP-based analysis enables ranking of feature importance and identification of dominant influencing factors of SVSM, while the SHAP beeswarm plots further reveal the direction and distribution of their impacts. Such interpretability facilitates both understanding of the prediction mechanism and control guidance. The main contributions of this paper are as follows:

1) A data-driven online probabilistic prediction framework for SVSM in AC/DC hybrid power systems is proposed. The framework explicitly accounts for system uncertainty and AC/DC coupling characteristics while ensuring real-time performance and interpretability.

2) A LightGBM–SHAP–KDE method is developed within this framework, combining the efficient learning capability of LightGBM, the interpretability provided by SHAP, and the probabilistic modeling strength of KDE, thereby enabling accurate, fast, and interpretable probabilistic SVSM prediction.

3) Extensive validation is carried out on representative test systems, including the IEEE 39-bus and the large-scale Polish 2383-bus winter-peak systems, under PMU/WAMS latency and asynchrony, topology changes, measurement noise, and missing data. Comparison with the Continuation Power Flow Method (CPF) and the Improved Cumulant Method (ICM) further confirms its effectiveness, scalability, and robustness.

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TABLE I
COMPARISON OF THE PROPOSED METHOD WITH PREVIOUS METHODS

Ref.	Evaluation method	System uncertainty	DC constraint	Real-time	Interpretability	Model Type
[2]-[7]	CPF/Thevenin-equivalent /Sensitivity/Modal analysis	--	--	--	√	**model-based**
[8]-[13]	ELM / CNNs/ BPNN/ ANFIS/ CD-RBF/ multi-CNN	--	--	√	--	**data-driven**
[14]-[19]	MCS/ Probabilistic analysis / Interval analysis	√	--	√	--	**data-driven**
[20]-[21]	Probabilistic CPF/ BLOPF	√	√	--	√	**model-based**
[22]-[25]	PPF-DKDE/Galerkin/ CNN–LSTM / BDLSTM	√	√	√	--	**data-driven**
This paper	LightGBM-SHAP-KDE	√	√	√	√	**data-driven**

∗ √ : The item is considered. –: The item is not considered

II. PROBLEM FORMULATION

A. *Static Voltage Stability Margin Index*

The problem of static voltage stability in power system is mainly caused by the system's inability to provide sufficient load active power or reactive power. At present, the mechanism analysis of static voltage instability is usually analyzed by P-V curve, which describes the corresponding relationship between bus voltage and load active power. The schematic diagram of the P-V curve is shown in Fig 1.

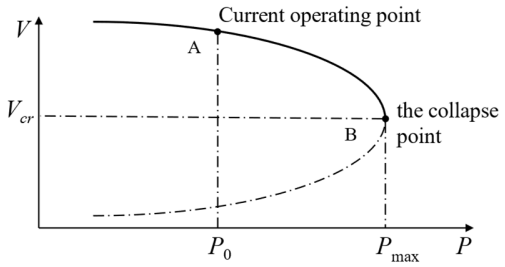


Fig. 1. Schematic of PV Curve.

As shown in Fig.1, the system voltage remains stable when the active power is below P_{max}, whereas it collapses once the power exceeds P_{max}. In this study, the SVSM is defined as the index to characterize system voltage stability and is adopted as the prediction target of the subsequent regression problem. The SVSM is expressed as follows:

$$SVSM = \frac{P_{max}}{P_0} \tag{1}$$

where P_0 is the active power of the load at operating point A of a certain power system; P_{max} is the maximum load active power representing the voltage collapse point B.

B. *Impact of Renewable Energy Uncertainty on SVSM*

In traditional power systems, uncertainty arises primarily from load variation. However, in grids with high renewable penetration, the stochastic output of wind and photovoltaic generation introduces additional uncertainty, resulting in greater SVSM, as illustrated in Fig. 2.

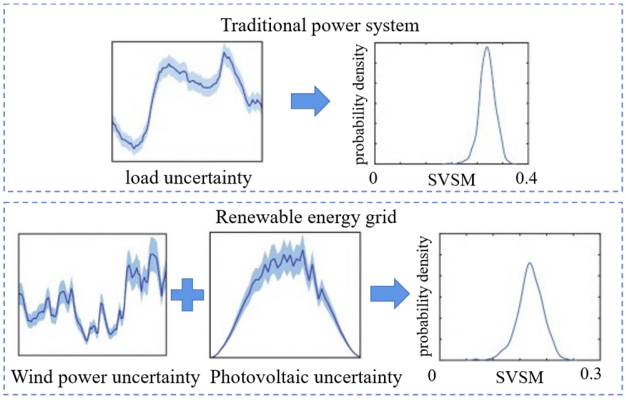


Fig. 2. The impact of uncertainty in renewable energy on SVSM.

As shown in Fig. 2, when only load uncertainty is considered, the probability distribution of SVSM is relatively narrow. Once renewable uncertainty is included, the distribution becomes significantly wider, and SVSM fluctuations increase markedly. Hence, it is necessary to predict the probability density distribution of SVSM by explicitly accounting for renewable volatility.

C. *Impact of VSC-HVDC on SVSM*

In traditional AC power grids, the evaluation of the SVSM is primarily based on power flow equations, where the system's tolerance to load disturbances is determined by constructing bus P–V curves. However, with the widespread integration of VSC-HVDC into power systems, both power flow distribution and operational constraints have undergone significant changes, making conventional SVSM evaluation methods inadequate for the requirements of AC/DC hybrid systems [26].

For AC/DC hybrid power systems with VSC-HVDC, additional constraints must be considered, including DC network power balance, the operational characteristics of VSC devices, and the variation of VSC converter operating states during SVSM calculation.

In this paper, a SVSM evaluation model for AC/DC hybrid power systems is developed by comprehensively incorporating these constraints. The power balance constraints of the AC/DC hybrid network are expressed in (2):

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$$\begin{cases} P_i - V_i \sum_{j=1} V_j\left(G_{ij}\cos\theta_{ij} + B_{ij}\sin\theta_{ij}\right) = 0 \\ Q_i - V_i \sum_{j=1}^{n} V_j\left(G_{ij}\sin\theta_{ij} - B_{ij}\cos\theta_{ij}\right) = 0 \\ P_m^{\mathrm{dc}} = V_m^{\mathrm{dc}} \sum_{n\in N_{\mathrm{dc}}} \frac{V_m^{\mathrm{dc}} - V_n^{\mathrm{dc}}}{R_{mn}} \\ P_m^2 + Q_m^2 \le S_{m,\max}^2 \\ V_m^{\mathrm{ac}} \in [V_{\min}, V_{\max}], V_m^{\mathrm{dc}} \in [V_{\min^{\mathrm{dc}}}, V_{\max^{\mathrm{dc}}}] \end{cases} \tag{2}$$

where P_i and Q_i are the active and reactive power of node i; Vi and V_j are the voltage amplitude of bus i and j; G_{ij} and B_{ij} are the real and imaginary part of admittance between bus i and j; P_m^{dc} and V_m^{dc} are the active power injected into the DC network by VSC converter m and the connected DC bus voltage; P_m and Q_m are the active and reactive power output of converter m on the AC side; V_m^{ac} is the AC side bus voltage connected to converter m; R_{mn} is the equivalent resistance of the DC line from m to n.

For the VSC-HVDC system, it is also necessary to consider the operational characteristics constraints and control methods of the VSC converter station, as shown in (3) and Table II:

$$\begin{cases} P_s = -U_s^2 G_{tf} + U_s U_f\left[G_{tf}\cos\Delta\delta_{sf} + B_{tf}\sin\Delta\delta_{sf}\right] \\ Q_s = U_s^2 G_{tf} + U_s U_f\left[G_{tf}\cos\Delta\delta_{sf} - B_{tf}\sin\Delta\delta_{sf}\right] \\ P_c = U_c^2 G_c - U_c U_f\left[G_c\cos\Delta\delta_{fc} - B_c\sin\Delta\delta_{fc}\right] \\ Q_c = -U_c^2 B_c + U_c U_f\left[G_c\sin\Delta\delta_{fc} + B_c\cos\Delta\delta_{fc}\right] \\ Q_f = -U_f^2 B_f \\ P_{\mathrm{loss}} = a + bI_c + cI_c^2 \end{cases} \tag{3}$$

where P_s, Q_s are active and reactive power injected at the AC system side; P_c, Q_c are active and reactive power at the VSC converter side; Q_f is reactive power of the filter; P_{loss} is power loss of the VSC converter; $U_s, \Delta\delta_s$ are voltage magnitude and phase angle of the AC system bus; $U_f, \Delta\delta_f$ are voltage magnitude and phase angle at the filter bus; $U_c, \Delta\delta_c$ voltage magnitude and phase angle of the VSC converter internal bus; $\Delta\delta_{sf}$ is phase angle difference between AC bus and filter bus; $\Delta\delta_{fc}$ is phase angle difference between filter bus and VSC converter; G_c, B_c are VSC converter admittance parameters; B_f is susceptance of the filter branch; I_c is VSC converter current; a, b, c are loss coefficients of the VSC converter.

TABLE II
CONTROL MODE OF VSC-HVDC

Control mode	DC control mode	Node type	AC control method	Node type
1	Constant DC voltage	V	Constant AC voltage	PV
2	Constant DC voltage	V	Constant reactive power	PQ
3	Constant active power	P	Constant AC voltage	PV
4	Constant active power	P	Constant reactive power	PQ

As shown in Table II, four common control modes of VSC-HVDC are summarized. In this paper, the SVSM calculation is carried out mainly under the control scheme where the DC side adopts V-P control mode and the AC side adopts the PV–PQ control mode.

In addition, the modified CPF method is employed in this paper to compute the SVSM of the system, which requires a continuous increase in the active power of all loads. Accordingly, the DC current reference must also rise with the load demand but remain below the maximum limit specified by the operational constraints of the VSC devices. Therefore, in addition to satisfying the constraints in (2) and (3), the following condition in (4) must also be enforced:

$$\begin{cases} I_{\mathrm{Vi}} = I_{\mathrm{Vi,ref}} - I_b \\ I_{\mathrm{Vi,ref}} = I_{\mathrm{Vi,ref\,0}} + \lambda b_{\mathrm{VSC}} \\ \left(I_{\mathrm{Vi},max} - I_{\mathrm{Vi}}\right) I_b = 0, I_b \ge 0 \\ I_{\mathrm{Vi}} \le I_{\mathrm{Vi},max} \end{cases} \tag{4}$$

where I_{Vi} the DC current of the VSC converter station connected to node i; $I_{\mathrm{Vi,ref\,0}}$ is the reference value of DC current at the ground state load level; $I_{\mathrm{Vi,ref}}$ is the reference value of DC current after the increase of load level; I_b is an auxiliary variable used to reflect whether the DC current of the VSC converter station has reached the upper bound; b_{VSC} is the direction of DC current growth, generally taking $I_{\mathrm{Vi,ref\,0}}$; λ is the multiple of load growth, which is SVSM.

III. ONLINE PREDICTION METHOD OF SVSM PROBABILITY

Before presenting the proposed online probabilistic prediction method for SVSM, the modeling of uncertainty scenarios of wind and solar power is first introduced to demonstrate how the uncertainty of renewable generation is incorporated. The detailed implementation of the LightGBM–SHAP–KDE algorithm is then described.

A. Scenario Modelling of Wind–Solar and Load Uncertainty

To accurately capture the fluctuation characteristics and correlation of wind and photovoltaic (PV) outputs, a joint wind–PV generation output fluctuation model is developed [27], [28]. The method consists of three main steps: marginal distribution modeling, Copula function coupling, and Latin hypercube sampling (LHS). This framework enables the generation of renewable scenarios with statistical consistency and representativeness.

In the first step, marginal distribution models are constructed for wind speed and solar irradiance. Assuming that the wind speed variable u follows a two-dimensional Weibull distribution, its probability density function is defined as follows:

$$f(v) = \left(\frac{k}{c}\right)\left(\frac{v}{c}\right)^{k-1} exp\left[-\left(\frac{v}{c}\right)^{k}\right] \tag{5}$$

where v is the wind speed; k and c is the two parameters of the Weibull distribution.

For the normalized solar irradiation intensity $s \in (0,1)$, it is assumed that it obeys the Beta distribution:

$$f_{\mathrm{PV}}(s) = \frac{\Gamma(\alpha+\beta)}{\Gamma(\alpha)\cdot\Gamma(\beta)}\left(\frac{s}{s_{max}}\right)^{\alpha-1}\left(1-\frac{s}{s_{max}}\right)^{\beta-1} \tag{6}$$

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where s is the actual solar irradiation; s_{max} is the maximum illumination intensity; α, β is the two parameters of the Beta distribution.

For load uncertainty, various probability distributions may arise depending on factors such as residential consumption behavior, load composition, and electricity pricing. Since the focus of this study is the generation and prediction of SVSM samples rather than reconstructing a detailed load forecasting model, a simplified yet reasonable modeling approach is adopted. The system under study is assumed to be a highly aggregated system, where numerous heterogeneous individual loads are represented collectively as a single equivalent system load. In such systems, short-term load variations can be approximated by a normal distribution according to the central limit theorem. The load uncertainty model adopted in this paper is formulated as follows:

$$f(x \mid \mu, \sigma^2) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \tag{7}$$

where μ is the mean of a normal distribution; σ^2 is variance; σ is standard deviation; x is random variablez.

To model the correlation between wind speed and solar irradiance, a Copula-based approach is adopted to construct the joint distribution. Let $F_v(v)$ and $F_s(s)$ denote the marginal cumulative distribution functions (CDFs); the joint probability density function of wind-solar output can then be expressed as:

$$f_{v,s}(v,s) = c(F_v(v), F_s(s); \theta) \cdot f_v(v) \cdot f_s(s) \tag{8}$$

where c is a two-dimensional Copula function, and θ is the correlation parameter.

In this paper, Gaussian Copula is used to construct a wind-light joint model [29]. Let the corresponding covariance matrix be Σ, then the form of Gaussian Copula is:

$$C(u, v; \rho) = \phi_{i_\Sigma}\left(\phi_i^{-1}(u), \phi_i^{-1}(v)\right) \tag{9}$$

where $\phi_i^{-1}(\cdot)$ is the inverse function of the standard normal distribution, and $\phi_{i_\Sigma}(\cdot)$ is the two-dimensional normal distribution function with a covariance matrix of Σ.

The correlation coefficient ρ in the Gaussian Copula is fitted by the maximum likelihood estimation (EM) method [30]:

$$\hat{\theta} = \arg\max_{\theta} \sum_{i=1}^{N} \log c(F_v(v_i), F_s(s_i); \theta) \tag{10}$$

Since different types of Copula functions exhibit varying abilities in capturing tail dependence, an appropriate Copula must be selected according to the characteristics of the application scenario. Therefore, this study considers four widely used Copula functions—Gumbel, Clayton, Gaussian, and Frank—for comparative analysis.

To identify the most suitable Copula for accurately modeling wind–solar output correlation, four performance metrics are employed: Kendall's rank correlation coefficient ρ_k, the Akaike information criterion (AIC), the squared Euclidean distance d_s, and the maximum deviation distance d_{max}.

Kendall's ρ_k measures the strength of dependence between two variables based on the rank correlation of all data pairs, providing a nonparametric dependence metric. A Copula model is considered more accurate when its theoretical value ρ_k is closer to the empirical value ρ_{k_emp}. The formulation is expressed as follows:

$$\begin{cases} \rho k = \frac{2(a-b)}{N(N-1)} \\ \rho k_{emp} = \frac{2}{N(N-1)} \sum_{i<j} \text{sign}\left[(x_i - x_j)(y_i - y_j)\right] \end{cases} \tag{11}$$

where a is the logarithm of consistency; b is the logarithm of inconsistency; N is the number of sampling points; (x_i, y_i) and (x_i, y_i) are sample points.

The AIC metric jointly considers the goodness of fit and model complexity; a lower value indicates better model performance and effectively prevents overfitting. Its formulation is given as follows:

$$AIC = 2K - 2\ln(L) \tag{12}$$

where K is the number of parameters in the model, which also represents the complexity of the model; L is the maximum likelihood estimate of the model.

The squared Euclidean distance d_s represents the sum of squared distances between two points and is used to quantify the discrepancy between the empirical data and the data generated by the Copula model. The maximum deviation distance d_{max} measures the worst-case divergence between the modeled and empirical distributions. The definitions of d_s and d_{max} are expressed as follows:

$$\begin{cases} d_s = \sum_{i=1}^{N} |C_1(u_i, v_i) - C(u_i, v_i)|^2 \\ d_{max} = \max\{|C_1(u_i, v_i) - C(u_i, v_i)|\} \end{cases} \tag{13}$$

where $C(u_i, v_i)$ is the joint distribution for Copula function; $C_1(u_i, v_i)$ is the empirical distribution of the sample.

In constructing the wind–solar fluctuation scenarios, wind speed variations are modeled using a Weibull distribution with a cut-in speed of 3 m/s, a rated speed of 13 m/s, a cut-out speed of 20 m/s, and parameters λ=7 and k=2. Solar irradiance variations are modeled using a Beta distribution with shape parameters α=2 and β=5. A total of 10,000 wind–solar samples are generated, and the joint distribution is established using four Copula models. The optimal Copula parameters are estimated via the EM algorithm. Fig. 3 show the 3D joint density surfaces of the Gaussian, Frank, Clayton, and Gumbel Copulas, respectively, while Table III compares the four performance metrics across the candidate models.

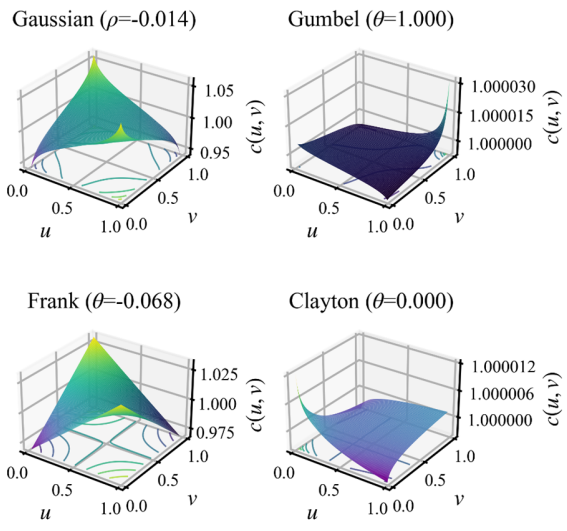


Fig. 3. The 3D joint density surfaces of four Copula models.

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The comparison of the four performance metrics for each Copula model is summarized in Table III. Among them, ρk denotes the theoretical Kendall rank correlation coefficient, and a value closer to the ρk_{emp} indicates better dependence fitting. For the remaining three metrics, smaller values correspond to better performance.

TABLE III
COMPARISON OF COPULA PERFORMANCE METRICS

Copula type	ρk	AIC	d_s	d_{max}
Gaussian	-0.0088	-0.8985	0.0001	0.0050
Gumbel	0.0000	2.0011	0.0003	0.0058
Clayton	0.0000	2.0007	0.0003	0.0058
Frank	-0.0076	0.0684	0.0001	0.0049

∗ The ρk_{emp} =-0.0076.

As shown in Table III, the Gumbel and Clayton Copulas yield larger AIC values and larger d_s and d_{max}, indicating weaker overall fitting performance despite their ability to capture upper-and lower-tail dependence. The Frank and Gaussian Copulas exhibit similar d_s and d_{max} values; however, the theoretical Kendall's ρk of the Frank Copula is closer to the −0.0076, suggesting slightly better dependence fitting. Nonetheless, its AIC is higher than that of the Gaussian Copula, indicating inferior model stability. The Gaussian Copula achieves the lowest AIC among all candidates, demonstrating the best likelihood performance with the least model complexity when modeling the joint dependence between wind speed and solar irradiance. Therefore, the Gaussian Copula is selected as the final model for capturing wind–solar correlation.

B. LightGBM-SHAP-KDE Method

To address the challenges posed by high-dimensional features to both the prediction accuracy and interpretability of SVSM, a joint LightGBM–SHAP–KDE framework is introduced.

LightGBM is a high-performance ensemble tree-based algorithm that achieves excellent training speed, accuracy, and scalability for large datasets. It has been widely applied in industrial practice [31] and is particularly suitable for complex power system scenarios considered in this study. For a training sample set $D(xi, yi)(i = 1,2, \dots, n)$, the LightGBM learning model can be formulated as follows:

$$f_0(x) = \underset{c}{argmin} \sum_{i=1}^{n} L(y_i, c) \tag{14}$$

where $L(y_i, c)$ is the loss function of the predicted value of the sample; c is the initial fitting value (usually the mean of all sample label values); n is the sample size.

LightGBM constructs a new decision tree at each iteration to minimize the sample loss. To enhance efficiency, it updates the model using residuals and assigns higher weights to samples with larger residuals, thereby focusing on those that are harder to fit. This mechanism enables LightGBM to achieve faster training compared with many other algorithms. The residuals at the t-th iteration are computed as follows:

$$r_{ti} = -\left\{\frac{\partial L[y_i, f(x_i)]}{\partial f(x_i)}\right\}_{f(x)=f_{t-1}(x)} \tag{15}$$

where $L[y_i, f(x_i)]$ is the loss function of the t-th iteration; $f(x_i)$ is the i-th learner of the LightGBM model and also the predicted value of the sample x_i.

According to the residual r_{ti}, the weight w_i of $f(x_i)$ can be calculated. The formula is as follows:

$$w_i = arg\ min_w \sum_{i=1}^{T} [r_{ti} - f_i(x, w_i)]^2 \tag{16}$$

SHAP is a game-theoretic approach for interpreting machine learning models. By calculating Shapley values, it quantifies the contribution of each feature to the prediction, providing a consistent and theoretically rigorous measure of feature importance. SHAP supports both global and local interpretability, facilitating the understanding of model behavior and assisting in feature reduction [32].

$$\phi_i = \sum_{S \in N(i)} \frac{|S|!(|N|-|S|-1)!}{|N|!} [f(S \cup \{i\}) - f(S)] \tag{17}$$

where ϕ_i represents the SHAP value of feature i, which quantifies its contribution to the model output; N is the set of all input features, and S is any subset of features that does not include i; $f(S)$ is the model prediction when only the features in set S are present, while $f(S \cup \{i\})$ indicates the prediction when feature i is added to S.

After obtaining the marginal contribution (SHAP value) of feature i to the prediction, the model output can be represented as the weighted sum of the SHAP values across all features, as expressed by:

$$f(x) = \phi_0 + \sum_{i=1}^{M} \phi_i \tag{18}$$

where ϕ_0 is the benchmark value of SHAP (usually the average output of the training set; $f(x)$ is the predicted value of the model for sample x.

KDE is a nonparametric technique for estimating the probability density function of a random variable. Unlike histograms, KDE yields smooth and continuous curves that more accurately represent the data distribution. It assigns weights to each data point through a kernel function and overlays them to form the overall density. Since KDE does not assume a predefined distribution, it is well suited for modeling uncertainties in power systems [33]. The formula is as follows:

$$\hat{f}_h(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{x - x_i}{h}\right) \tag{19}$$

where h is the bandwidth; x is the stable margin sample point to be fitted; K is the kernel function.

In this paper, a Gaussian kernel function is employed for density estimation, formulated as follows:

$$K(v) = \frac{1}{\sqrt{2\pi}} exp\left(-\frac{1}{2} v^2\right) \tag{20}$$

where v is the standardized distance between the data point and the evaluation point.

C. Online Probabilistic SVSM Estimation Procedure

The overall workflow of the proposed LightGBM–SHAP–KDE method is illustrated in Fig. 4, and the detailed computational procedure is summarized in Algorithm 1.

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Algorithm 1: The LightGBM-SHAP-KDE Algorithm

Inputs: Historical WAMS/PMU dataset D.
Initialize actor and number of selected features for each agent, and memory buffer.
Output: Trained LightGBM model M, SHAP-ranked features S, SVSM values Y and KDE distribution F.
for each epoch **do**
    t = 0: set up random fluctuation scenarios for generators and loads, and initialize the power system state.
    for each calculation step **do**
        While t < total simulation time
            Train initial LightGBM model M_0 on D.
            Compute SHAP values for all features in D using M_0 and rank features.
            Construct reduced dataset X using top-m features in S.
            Retrain LightGBM model M on X → Y.
            $t = t + 1$
        end while
        Predict SVSM values Y = M(X).
        Obtain the probability distribution F by fitting the KDE model on Y.
    end for
    Analyze Y results based on SHAP method.
    Provide decision support and control guidance based on SHAP analysis.
end for

Algorithm 1 presents the pseudocode of the proposed LightGBM–SHAP–KDE framework for probabilistic online estimation of SVSM. The algorithm is designed to be adaptive: when a new operating scenario arises, the SHAP method is employed to re-rank and reduce the input features, thereby selecting an appropriate training dataset for the LightGBM model. The LightGBM predictions are then fitted into a probability distribution using KDE, while SHAP analysis further explains the positive and negative impacts of input variables on the predicted results. This process ensures accurate, probabilistic, and interpretable estimation of SVSM.

D. Alignment Strategy of Data Latency and Asynchrony

According to the IEEE Standard for Synchrophasor Data Transfer for Power Systems (C37.118.2–2011), typical values for the transmission delay between PMU and Phasor Data Concentrator (PDC) are between 20 ms to 50 ms, and this delay may be further increased due to temporary data congestion over the communication links. To ensure that the proposed SVSM prediction framework remains timely and reliable under such conditions, appropriate strategies have been adopted in both data preprocessing and model training to address the issues of asynchronous data arrival and temporal latency. The specific details are as follows:

1) Sliding time window alignment strategy: To cope with asynchronous PMU arrivals, a sliding time-window alignment strategy is introduced. The sliding window allows input data to contain a certain degree of latency and asynchronous. As long as the delay is within a predefined interval Δt, the algorithm selects the data closest to the target prediction time. Assume that the window length is Δt and the actual prediction instant is t_k. For a prediction instant t_k, the interval $[t_k - \Delta t,\ t_k + \Delta t]$ is searched for each PMU channel to locate the nearest sample. Missing channels are completed using local interpolation. Because that typical PMU–PDC delays range from 20 to 50 ms, the sliding time-window length Δt is set to approximately two to four times the upper bound of this delay (100-200ms).

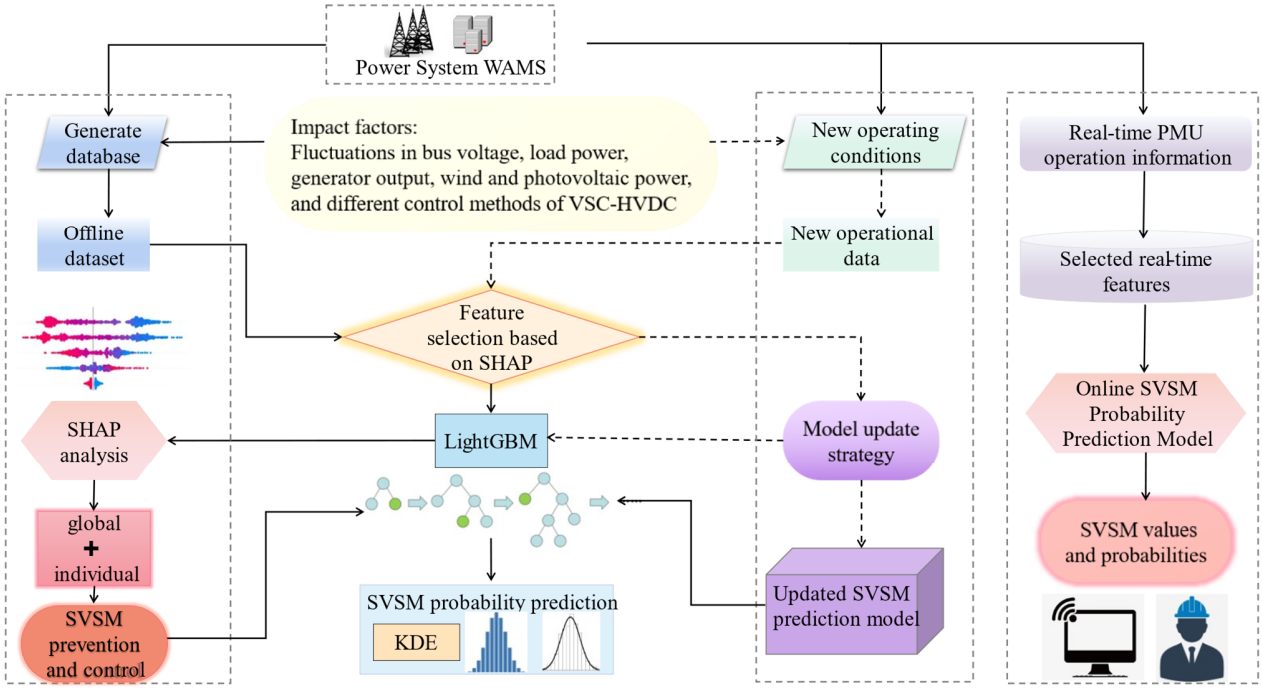


Fig. 4. Flowchart of online probabilistic estimation for SVSM.

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This choice allows the window to accommodate most random timing jitters while remaining compatible with the 1–5 s refresh cycle of practical EMS applications. The process is expressed mathematically as:

$$\begin{cases} x_i(t_k) = x_i(t_i^*) \\ t_i^* = min|t - t_k| \end{cases} \tag{21}$$

where $t\epsilon[t_k - \Delta t,\ t_k + \Delta t]$; $x_i(t_k)$ is the input vector of the model at time t_k.

2) Data-consistency verification strategy: To prevent unreliable predictions caused by severely desynchronized or incomplete measurements, a data consistency verification process is established before each SVSM prediction cycle. Assume that η_{sync} represent the proportion of time aligned channels in the window, and η_{miss} represent the feature deletion rate. Based on typical WAMS engineering experience, the minimum acceptable time-alignment ratio is set to η_{sync}= 0.95, while the maximum allowable feature-missing rate is limited to η_{miss} = 0.3. If the data fail to satisfy these requirements, the current data frame is classified as low-confidence, and the previous valid prediction is reused while a data-quality warning is issued. The formulation is expressed as follows:

$$\begin{cases} \eta_{\text{sync}} = \frac{1}{M}\sum_{i=1}^{M} 1\left(\delta_i \le \Delta t_{\text{sync}}\right) \\ \eta_{\text{miss}} = m_{\text{miss}}/M \end{cases} \tag{22}$$

where $\delta_i = |t_i^{PMU} - t_k|$, which represents the arrival time deviation of the data of each PMU; Δt_{sync} is the sliding time window set in this paper. If $\delta_i \le \Delta t_{\text{sync}}$, it is considered that the PMU channel i is synchronous; m_{miss} and M are the number of missing features and the total number of input features.

3) Delay-aware training strategy: In real-time WAMS environments, the measurements available at time typically correspond to an earlier true system state due to communication latency. State-of-the-art delay-aware transient stability assessment (TSA) studies [34] have addressed this issue by training models with lagged input sequences, enabling the predictor to learn the inherent mapping from delayed observations to the current (or even future) stability condition. This principle is adopted here for SVSM prediction. The formulation is expressed as follows:

$$\begin{cases} y(t_k) = f\left(x(t_k - \tau)\right) \\ \tau = L \cdot \Delta t \end{cases} \tag{23}$$

where $y(t_k)$ is the SVSM predicted by the model at time t_k; τ is the total input delay time, where L is the number of delayed cycles and Δt is the prediction interval.

IV. Case Study

The modified IEEE 39-bus AC/DC power system is taken as an example, and its topology is illustrated in Fig. 5. In this system, the conventional generators at buses 37 and 38 are replaced by a wind turbine and a photovoltaic station, respectively. Both renewable units are modeled as PQ-type generators with rated active powers of 540 MW and 830 MW, operating under constant power factor strategies of 0.98 and 1.0. The AC line between buses 4 and 14 is substituted with a bipolar VSC-HVDC line, where bus 14 serves as the rectifier and bus 4 as the inverter. The DC voltage is specified as 500 kV, and the converters adopt three-level structures with modulation ratios ranging from 0.8 to 1.0. The per-unit values of phase resistance, reactance, and DC line resistance are 0.006, 0.07, and 0.01, respectively. The transmission capacity of the DC link is set to 335 MVA, consistent with the original AC line rating. For the VSC control strategy, a typical PV–PQ control mode is applied, with VSC1 controlling active power and AC voltage, and VSC2 controlling DC voltage and reactive power.

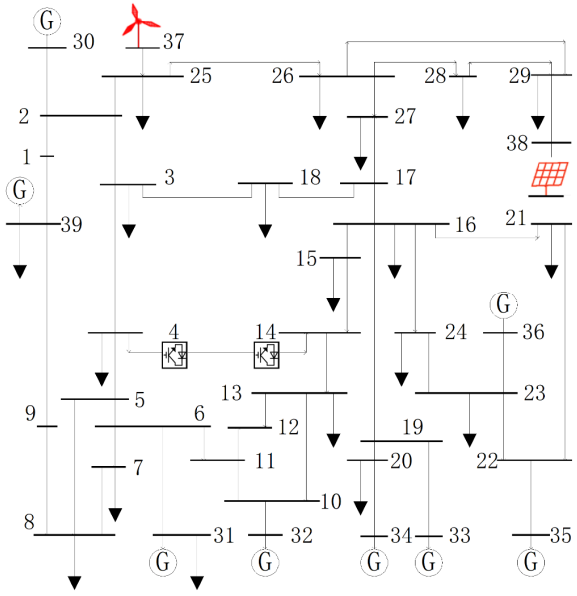
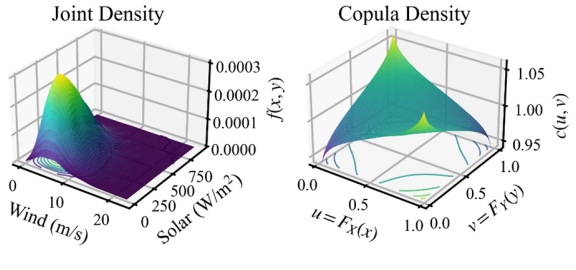


Fig. 5. Modified IEEE 39 bus AC/DC system in the case study.

A. SVSM Sample Generation under Wind–Solar and Load Uncertainties

In the wind–solar fluctuation scenarios considered in this study, the Gaussian Copula is identified as the most suitable model for capturing the dependence between wind and solar resources. Using this Copula, 10,000 samples of wind speed and solar irradiance fluctuations are generated. After converting these samples into power outputs through wind turbine and PV power conversion curves, Latin hypercube sampling (LHS) is applied, followed by K-means clustering. A total of 3,400 representative wind–solar output scenarios are obtained. Fig. 6(a) compares the joint probability density of wind–solar fluctuations with the corresponding Gaussian Copula density in the uniform domain, while Fig. 6(b) presents the frequency histogram of the resulting typical wind–solar output scenarios.



(a)

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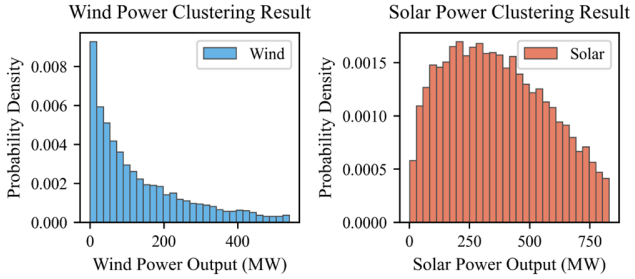


Fig. 6. (a) 3D surface plot of wind-solar probability density and Gaussion-Copular density.
(b) Histogram of output frequency of wind turbines and photovoltaic stations.

In highly aggregated power systems, short-term load variations are typically modeled as normal disturbances with magnitudes ranging from 10% to 20% of the nominal demand. To simulate load uncertainty in a manner consistent with practical operating conditions, a 15% normal perturbation is applied to the active power component of each load while maintaining a constant power factor. For illustration, Fig. 7 shows the frequency histogram of the load at Bus 3, whose nominal power is 322 MW, after applying a 15% normal fluctuation.

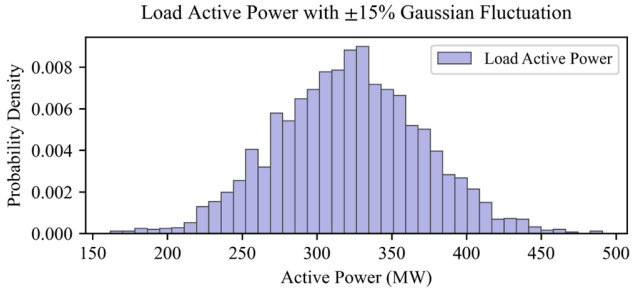


Fig. 7. Active power frequency histogram of load.

In addition, to better reflect real-world operating conditions, small normal perturbations are also introduced to bus voltages, phase angles, and synchronous generator outputs. In this study, these variations are set to 2%.

In this paper, Matpower and MatACDC were employed to model and analyze complex AC/DC hybrid networks. MatACDC is an open-source power flow analysis toolbox developed based on Matpower, which comprehensively incorporates AC/DC network constraints, VSC-HVDC operating constraints, and the multiple control mode constraints of DC devices.

In the modified IEEE 39-bus system, the sample generation process is described in detail. In this system, each SVSM sample consists of 144 input features, including bus voltage magnitudes and phase angles, active and reactive load powers, generator outputs, and the voltage and power measurements at both AC terminals of the VSC-HVDC link. The output variable is the SVSM. To generate training and validation samples, wind and photovoltaic outputs were produced using the Copula-based joint Weibull–Beta distribution and sampled via LHS. Based on the disturbance models introduced earlier, bus voltages and phase angles, generator active outputs were perturbed using small normal variations, whereas load powers were varied using a 15% normal fluctuation. System-level load growth was then applied under constant power factors, with the incremental demand distributed proportionally among conventional generators. Generator reactive power limits were enforced automatically by switching PV buses to PQ buses when limits were reached. For each perturbed operating point, an AC/DC hybrid power flow was solved using MatACDC, with the VSC-HVDC converters operating in V–P control on the DC side and PV–PQ control on the AC side. The resulting SVSM was obtained using a continuation power-flow method considering DC constraints, producing 3,000 training samples and 400 validation samples.

All simulations are performed on a personal computer equipped with an Intel Core i7 CPU of 2.7-GHz and 32 GB of RAM. The algorithms are implemented in Python 3.12, and the computation time is recorded using the timing utilities provided by Python's time module.

B. LightGBM-SHAP Point Prediction for SVSM

Prior to SVSM prediction, hyperparameter tuning of the LightGBM model is required. In this study, the Optuna framework is employed with the root mean square error (RMSE) as the optimization objective. As hyperparameter optimization is not the primary focus of this work, further details are omitted.

After determining the model parameters, redundancy analysis and dimensionality reduction of the input features are performed. The SHAP method is applied to rank the importance of all 144 features with respect to SVSM prediction and to identify the feature dimension corresponding to the minimum RMSE. The ranking results and the associated RMSE variation curve are illustrated in Fig. 8.

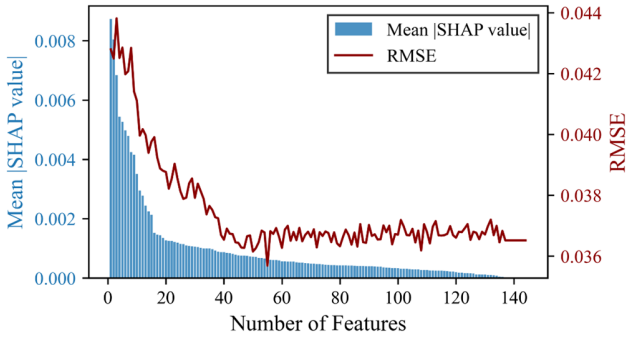


Fig. 8. Ranking of input features by SHAP and RMSE curve plot.

As shown in Fig. 8, the prediction RMSE of LightGBM reaches its minimum value of 0.0357 when the feature dimension is reduced to 55, corresponding to the best predictive performance. To further demonstrate the improvements of the LightGBM–SHAP method in terms of accuracy and computational efficiency, a comparison with the original LightGBM algorithm is conducted, and the results are summarized in Table IV.

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TABLE IV
COMPARISON OF PREDICTION SPEEDS OF FOUR ALGORITHMS

Algorithm name feature	Feature number	RMSE	Training time/s	Prediction time/s
LightGBM	144	0.0371	1.571	0.028
LightGBM-SHAP	55	0.0357	0.533	0.017

To validate the advantages of LightGBM–SHAP in prediction accuracy and computational efficiency, three widely used algorithms—XGBoost, SVM, and RF—were selected for comparison. The RMSE variations of the four algorithms after SHAP-based dimensionality reduction are illustrated in Fig. 9.

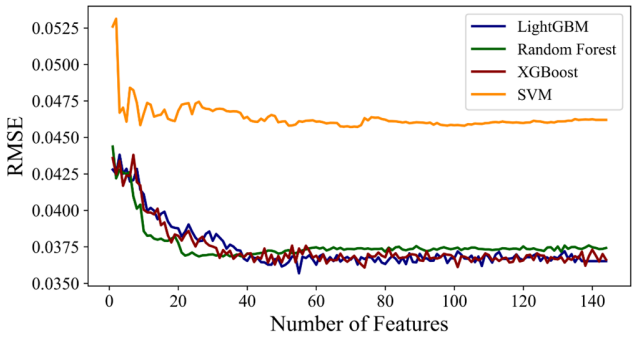


Fig. 9. Comparison of RMSE among four algorithms with varying numbers of features.

The time and accuracy of the four algorithms for predicting SVSM under the optimal feature number are shown in Table V:

TABLE V
COMPARISON OF PREDICTION SPEEDS OF FOUR ALGORITHMS

Algorithm name feature	Feature number	RMSE	Training time/s	Prediction time/s
LightGBM-SHAP	55	0.0357	0.533	0.017
XGBoost-SHAP	74	0.0361	1.587	0.021
RF-SHAP	39	0.0368	0.526	0.096
SVM-SHAP	70	0.0457	0.078	0.153

As shown in Fig. 9 and Table V, SVM yields the lowest prediction accuracy. RF performs relatively well when the feature dimension is small, but its accuracy declines once the number of features exceeds 39, becoming inferior to both XGBoost and LightGBM. Although LightGBM achieves accuracy comparable to XGBoost, it offers faster prediction speed and superior overall performance, making it more suitable for online SVSM prediction.

To further assess the prediction capability of the LightGBM model, Fig. 10 and Fig. 11 present the comparison between predicted and actual values, as well as the relative error distributions, based on 3,000 training samples and 400 validation samples.

Figs. 10 and 11 show that, for 3,000 training samples, the LightGBM–SHAP algorithm achieves a MAPE of 0.9%, and for 400 validation samples, the MAPE is 2.8%. The SHAP-based dimensionality reduction requires 1.625 s, while the LightGBM training and online prediction times are 0.533 s and 0.017s, demonstrating the high prediction accuracy and computational efficiency of the proposed method.

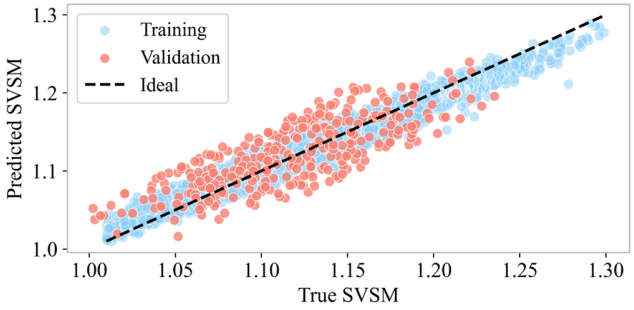


Fig. 10. Comparison of predicted and actual results.

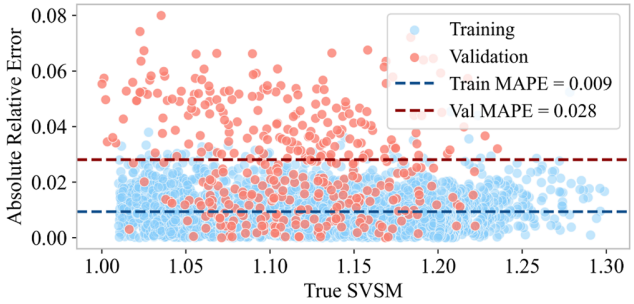


Fig. 11. Relative Errors in the Modified IEEE 39-Bus System.

C. KDE Estimation of SVSM Probability Distribution

After the SVSM values are predicted using the LightGBM–SHAP method, their probability distribution is estimated by KDE. In this study, a Gaussian kernel is employed, and the results are illustrated in Fig. 12.

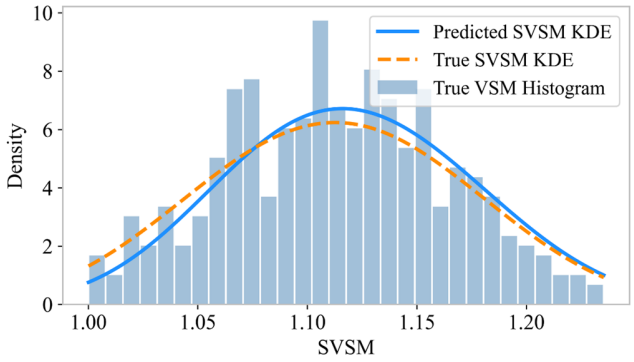


Fig. 12. SVSM probability distribution prediction based on KDE.

As shown in Fig. 12, the estimated probability distribution closely matches the actual SVSM distribution, indicating strong fitting capability. In addition, the KDE computation requires only 0.001 s, further demonstrating the real-time performance of the proposed approach.

To evaluate the impact of renewable penetration on SVSM prediction, three penetration levels of 10%, 20%, and 30% were considered. The probabilistic prediction results under these scenarios are shown in Fig. 13.

Fig. 13 shows that the SVSM primarily fluctuates between 1 and 1.25. As the renewable penetration level increases, the probability distribution of SVSM shifts leftward and exhibits greater variability. This occurs because the replacement of conventional generators with renewable units reduces the system's reactive power support capability, thereby weakening

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voltage stability, and higher penetration levels further exacerbate this deterioration.

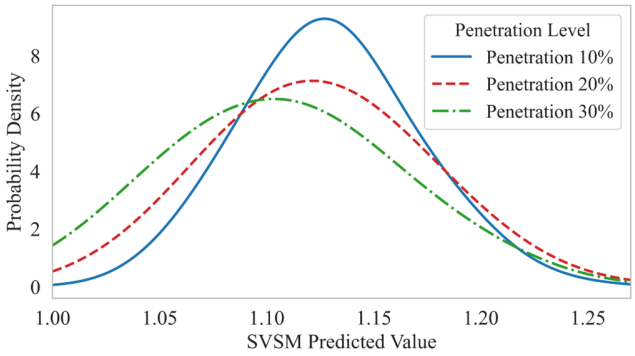


Fig. 13. Predicted probability distributions of SVSM under different renewable penetration levels.

D. Analysis of Influencing Factors based on SHAP Method

The SHAP method is employed to provide interpretability for the SVSM prediction results, including global analysis of feature importance ranking and directional effects, as well as local analysis of the impact of individual features on SVSM. The global results are illustrated in Figs. 14 and 15. Fig. 14 presents the importance ranking of the top ten features, obtained from the mean absolute SHAP values. Fig. 15 shows the beeswarm plot of these features, where positive and negative values indicate their favorable or adverse influence on the SVSM prediction. The magnitude of the SHAP value is reflected by the color depth, and the red-to-blue gradient corresponds to feature values decreasing from high to low.

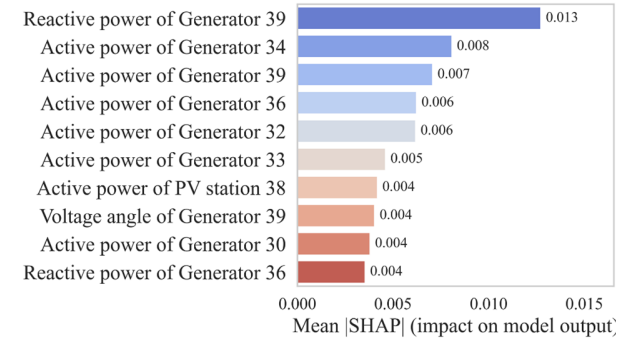


Fig. 14. Feature importance ranking obtained by shap analysis.

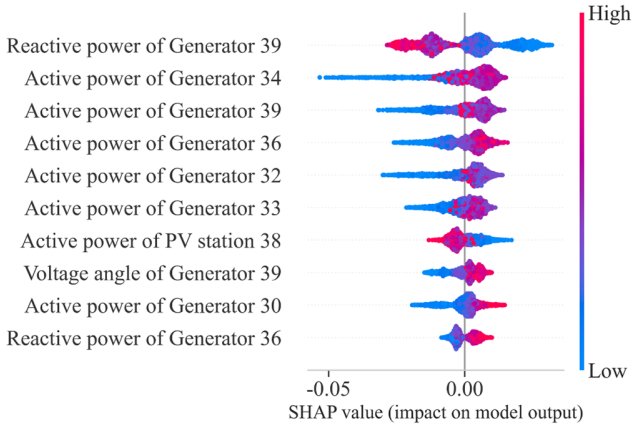


Fig. 15. SHAP Beeswarm plot of positive and negative feature impacts.

As shown in Fig. 15, among the top ten features, the reactive power of Generator 39 has the highest importance. The concentration of red points on the left side indicates that higher reactive power output from Generator 39 corresponds to lower predicted SVSM values. The second most important feature is the active power of Generator 34, where the accumulation of red points on the right side implies that greater active power output leads to higher SVSM values. Similar interpretations can be drawn for the remaining features.

To further illustrate individual feature effects, SHAP dependency plots are employed. For instance, the active power output of PV Station 38 is analyzed separately, and its dependency plot is presented in Fig. 16.

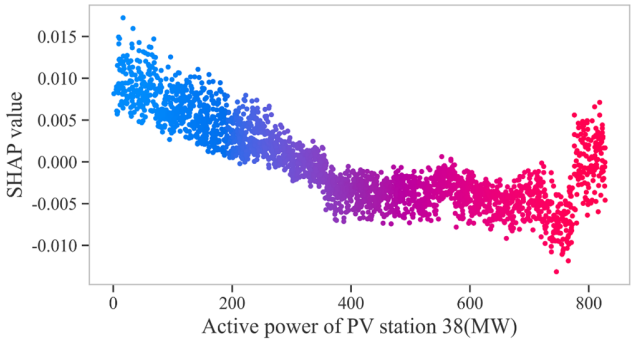


Fig. 16. SHAP dependency plot of the active power output of the PV.

As shown in Fig. 16, the blue points are concentrated in the 0–400 MW range, whereas the red points are clustered in the 400–800 MW range. Once the active power output of the PV station exceeds 400 MW, the corresponding SHAP value changes from positive to negative. This indicates that maintaining the PV output within the lower range of 0–400 MW in practical operation can enhance SVSM and thereby improve system voltage stability.

E. SHAP-Guided Decision Support and Control Adjustments

To illustrate the usefulness of the SHAP-based explanatory analysis, a simplified decision-support procedure is constructed using the most influential control variables identified from the SHAP rankings and dependency plots. The adjustment directions follow the positive or negative impact of each variable on the SVSM, and the ranges are selected according to the sensitivity revealed by the SHAP dependency curves. This module is intended only to demonstrate how SHAP-guided variable adjustments influence the SVSM, rather than to represent an actual EMS control strategy. The resulting effect of these SHAP-guided adjustments on the SVSM distribution is presented in Fig. 17.

As shown in Fig. 17, the SVSM probability distributions before and after applying the SHAP-guided adjustments exhibit a clear improvement in voltage stability. The control variables were selected from the top five features identified in the SHAP importance analysis, and their adjustment directions and magnitudes were determined using the corresponding SHAP dependency plots. In this study, the adjustments consisted of reducing the reactive power output of Generator 39 and moderately increasing the active power outputs of Generators 34, 39, 36, and 32, which collectively exert the most beneficial

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influence on the SVSM. Under 20% renewable penetration, the probability of SVSM falling below 1.1 decreases from 42.4% to 5.99%, and the 95% confidence interval narrows from [1.02, 1.18] to [1.11, 1.26]. These results indicate that SHAP-derived guidance can effectively enhance the SVSM and provide meaningful decision-support signals for preventive control.

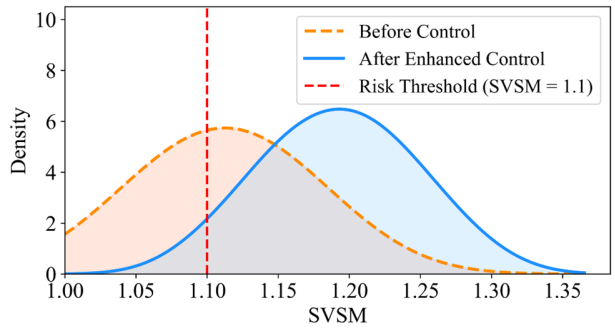


Fig. 17. SVSM probability distributions before and after adjustments.

It should be noted that the SHAP-based prioritization reflects the influence of variables on the SVSM only. Practical EMS operation must still determine the actual control actions by considering device capability limits, voltage and thermal security margins, generator excitation limits, and the economic impacts associated with reactive adjustments, tap changes, switching operations, and loss variations. SHAP helps by narrowing the optimization search space to the most influential and feasible control variables, and by providing a continuously updated, data-driven importance ranking that supports fast and reliable preventive control decisions under high renewable penetration.

F. Scalability Validation on Polish 2383- Bus Winter-Peak System

To evaluate the effectiveness of the proposed LightGBM–SHAP–KDE method in large-scale systems, the Polish 2383-bus winter-peak system was modified. The test system consists of 2383 buses, 2896 transmission lines, 327 generators, and 1817 loads, with a total peak demand of 24,558 MW. In this study, generators 17, 67, 127, 63, 131, and 176 were replaced by wind turbine units, while generators 1426, 1416, 911, 912, and 1192 were replaced by photovoltaic units. In addition, AC lines 15–165 and 31–6 were replaced by two VSC-HVDC transmission lines, transmitting 400 MW and 600 MW of active power, respectively, with 300 MW of reactive power, at a DC voltage level of 500 kV. The control mode of the two VSC-HVDC lines is V-P control mode on the DC side and PV-PQ control mode on the AC side. When the renewable penetration level was set to 30%, 3000 test samples and 700 validation samples were generated. The feature set contained 10,192 dimensions, including the voltage magnitude and phase angle of all buses, the active and reactive power of all loads and generators, and the voltage as well as active and reactive power of the two VSC-HVDC converter stations. After feature dimensionality reduction using the SHAP method, the optimal feature subset was reduced to 450 dimensions, as illustrated in Figs. 18–23.

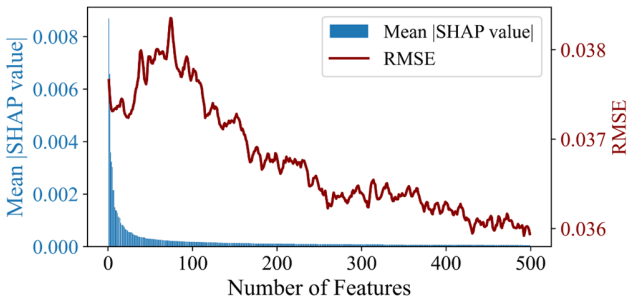


Fig.18. RMSE ranking of features after SHAP-based dimensionality reduction.

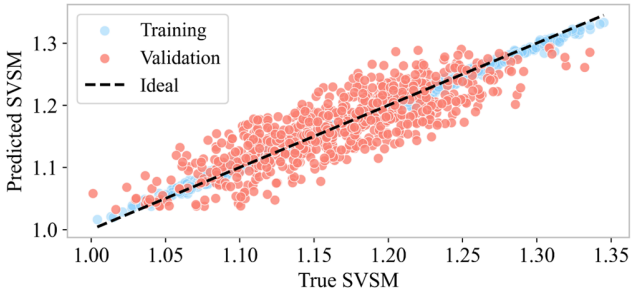


Fig. 19. Comparison of predicted and actual results.

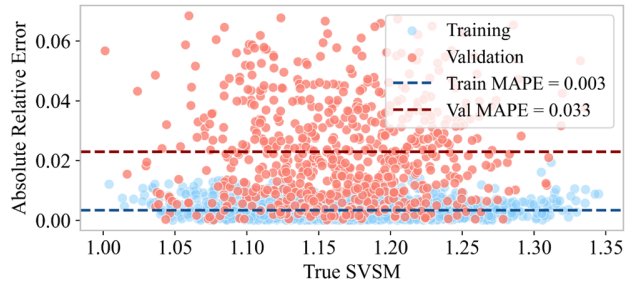


Fig. 20. Relative errors in the modified Polish 2383-bus system.

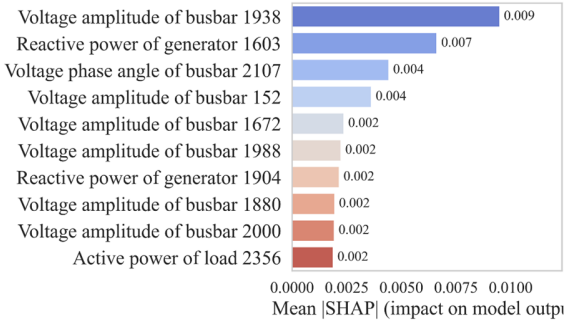


Fig. 21. Feature importance ranking obtained by shap analysis.

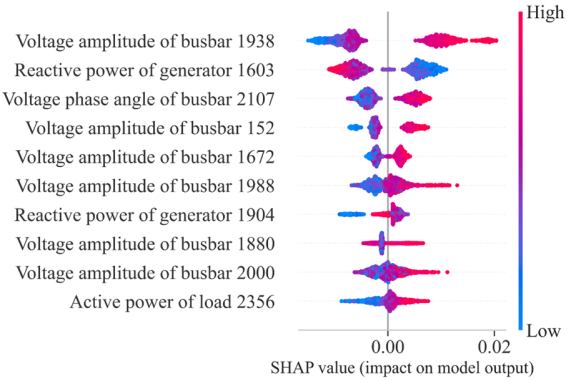


Fig. 22. SHAP Beeswarm plot of positive and negative feature impacts.

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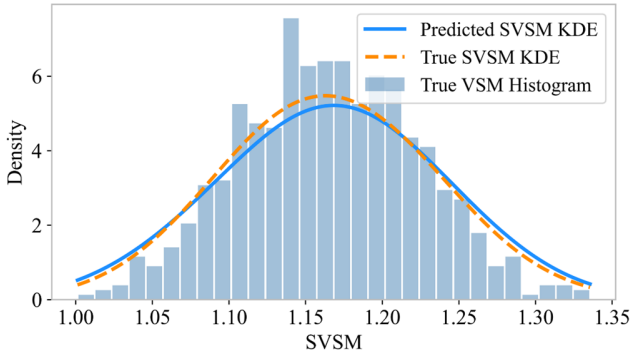


Fig. 23. SVSM probability distribution prediction based on KDE.

From Figs. 18–23, it can be observed that the proposed algorithm also delivers satisfactory predictive performance on the modified Polish 2383-bus winter-peak system. After dimensionality reduction to 450 features, the average relative prediction error reaches 3.3%. The LightGBM model requires 2.919 s for training and 0.671 s for prediction, the SHAP-based dimensionality reduction takes 5.675 s, and the KDE-based probabilistic fitting requires only 0.002 s. These results demonstrate that the proposed framework achieves both high prediction accuracy and excellent computational efficiency.

The SHAP ranking analysis further indicates that the bus voltages at buses 1938, 2107, and 1988 have the most significant influence on SVSM, implying that this region is particularly vulnerable to voltage instability. Generator 1603 and buses 1672 and 152, which are closely associated with the transmission corridors in this area, are also identified as highly influential variables. These findings suggest that system operators should pay particular attention to bus 1938 and its surrounding region, and may use the analysis presented in this study as decision-support guidance to enhance voltage stability and reduce the risk of voltage collapse.

Furthermore, SVSM prediction was also carried out for the modified Polish 2383-bus winter-peak system under renewable penetration levels of 10%, 20%, 30%, and 40%. The predicted probabilistic distributions of SVSM for these four scenarios are illustrated in Fig. 24.

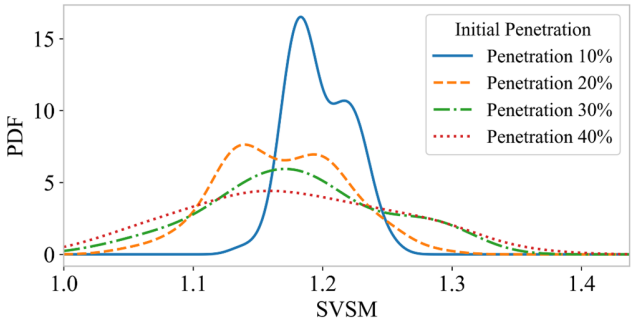


Fig. 24. Predicted probability distributions of SVSM under different renewable penetration levels.

From Fig. 24, it is observed that as the renewable penetration level increases from 10% to 40%, the volatility of the system SVSM significantly increases. Furthermore, compared with the modified IEEE 39-bus system, the SVSM distribution of the modified Polish 2383-bus winter-peak system exhibits a wider fluctuation range and more pronounced variations in curve shape. Specifically, the SVSM of the large-scale system fluctuates between 1.0 and 1.4, whereas that of the modified IEEE 39-bus system varies between 1.0 and 1.25. These results indicate that the large system possesses greater redundancy and stronger buffering capability, enabling it to accommodate a higher share of wind and solar integration without experiencing rapid voltage collapse.

G. Performance Comparison With Traditional Methods

To evaluate the advantages of the proposed LightGBM–SHAP–KDE algorithm over traditional model-based approaches, the Continuation Power Flow (CPF) method and the Improved Cumulant Method (ICM) are selected for comparison.

The CPF method is the most widely used classical technique for computing the SVSM. In the comparative evaluation, the CPF results are treated as the true value as well as the source of training samples. The ICM performs a local linearization around the operating point and analytically propagates the statistical moments of the input random variables. Crucially, the ICM incorporates high-order cumulants, specifically skewness and kurtosis, and utilizes the Cornish-Fisher expansion to construct the confidence intervals. This approach effectively allows for the accurate reconstruction of non-Gaussian probability distributions inherent in systems with renewable energy integration, thereby serving as a robust and rigorous baseline for comparison.

To further assess the probabilistic prediction performance of the different methods, the prediction interval coverage probability (PICP) and the prediction interval normalized average width (PINAW) are employed. The PICP reflects the proportion of true SVSM values that fall within the predicted interval, while the PINAW measures the average width of the interval. The corresponding formulas are given as follows:

$$\begin{cases} PICP = \frac{1}{N}\sum_{i=1}^{N} 1(\hat{y}_i^{\,L} < y_i < \hat{y}_i^{\,U}) \\ \text{PINAW} = \frac{1}{N}\sum_{t=1}^{N}(U_t - L_t)/(y_{max} - y_{min}) \end{cases} \tag{24}$$

where N is the total number of test samples; y_i is the true value; $\hat{y}_i^{\,L}$ and $\hat{y}_i^{\,U}$ are the lower and upper bounds of the predicted confidence interval; $1(\cdot)$ is the indicator function, with a value of 1 when the condition in parentheses is met, and 0 otherwise; U_t and L_t are the upper and lower bounds of the prediction interval at time t; y_{max} and y_{min} are the maximum and minimum values of the true SVSM in the entire sample.

The modified IEEE 39-bus system is used as the test case for the comparative evaluation of the three methods. For the modified Polish 2383-bus winter-peak system, the same workflow applies and the experiments are therefore not repeated.

Although classical probabilistic power flow studies based on the ICM typically do not account for temporal variations and are instead performed on static probability distributions, the comparative experiments in this study place particular emphasis on real-time applicability. Consequently, it is necessary to incorporate time-series modeling.

First, it is necessary to introduce temporal correlation into the joint probabilistic model of wind speed and solar irradiance constructed using the Gaussian Copula. Therefore, this paper employs a simple Autoregressive Model of Order 1 (AR (1)) to generate time series samples. The AR (1) process describes the

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time series evolution by relating the current value linearly to its immediate predecessor and a white noise term. Since this standard technique is not the core focus of this study, it is not discussed in further detail. The resulting 2-hour wind–solar power time series (1-s resolution) preserves the marginal distributions, dependence structure, and temporal autocorrelation of the Copula model, as shown in Fig. 25.

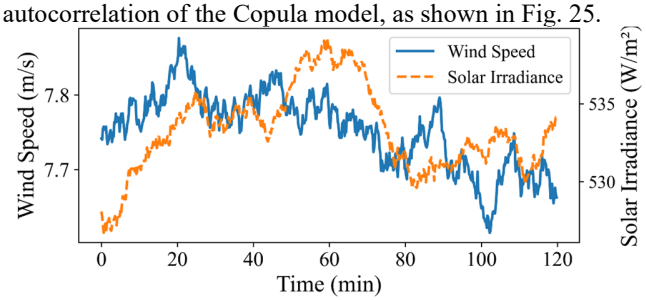


Fig. 25. Typical daytime wind and solar time series fluctuation curve.

In addition, to characterize short-term load uncertainty, a Gaussian AR (1) disturbance with a standard deviation of 15% is superimposed on the nominal active power of all 19 load buses, while maintaining constant power factors. Fig. 26 shows the resulting 2-hour load fluctuation trajectory at Bus 3.

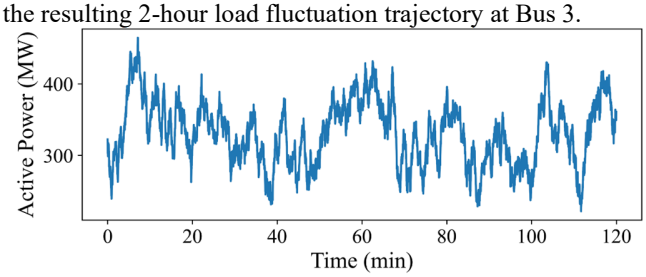


Fig. 26. Active power fluctuation time series at Bus 3.

Subsequently, the SVSM at each time instant within the simulated period is computed using the CPF method, from which 3,600 samples are selected for training and 600 samples for validation.

Since the KDE component of the proposed LightGBM–SHAP–KDE framework requires a batch of samples within a given interval to estimate the SVSM confidence bounds, an appropriate temporal window must be defined so that KDE estimation can be performed periodically. Considering that practical power system dispatch cycles typically range from 5 to 15 minutes, a 5-minute window is adopted in this study, with the confidence level set to 95%. The resulting estimation performance is illustrated in Fig. 27.

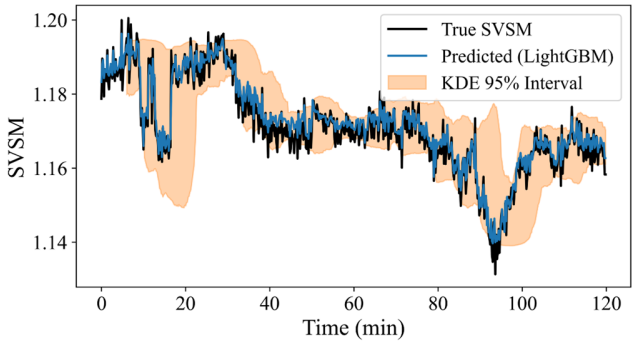


Fig. 27. The interval prediction situation of the method in this article.

For probabilistic SVSM interval estimation, this study employs the ICM, in which the statistical moments of wind power, photovoltaic generation, and load uncertainties are used as input parameters. Since the comparative analysis in this work focuses on real-time applicability, a temporal window is introduced to ensure a consistent evaluation framework across methods. Accordingly, a 5-minute window and a 95 percent confidence level are adopted. The corresponding results obtained using the ICM are presented in Fig. 28.

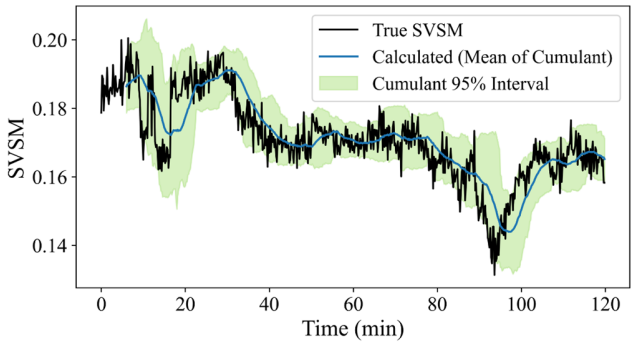


Fig. 28. Interval calculation of Cumulant method.

A comprehensive comparison of the proposed method and the two traditional model-based approaches is summarized in Table VI. The time metric represents the total computation or prediction time for all validation samples. The CPF results are treated as the ture SVSM values and reference benchmark.

TABLE VI
COMPARISON OF THREE METHODS FOR EVALUATING SVSM

Algorithm	PICP	PINAW	Time/s
CPF	1	0.251	744.625
ICM	0.965	0.278	0.523
LightGBM-SHAP-KDE	0.953	0.266	0.167

From Figs. 26–27 and Table VI, it can be observed that the proposed method achieves an online evaluation time of only 0.167 s, making it the fastest among the three approaches and highly suitable for real-time applications. The CPF method, serving as the benchmark for true SVSM values, requires the longest computation time and is restricted to offline analysis. Regarding the ICM, although it attains a 96.5% interval coverage rate by incorporating high-order statistics to handle non-Gaussianity, its reliance on linearized sensitivity leads to substantially wider prediction intervals (higher PINAW). In contrast, the proposed method delivers both accurate point predictions and high-quality interval estimates, achieving 95% coverage with significantly narrower interval widths. These results demonstrate that, compared with traditional model-based techniques, the proposed approach offers superior performance in interval-estimation quality and real-time computational efficiency.

Additionally, since the comparison focuses on the online applicability of different methods, the reported computation time for the proposed approach includes only the online LightGBM point-prediction time and the KDE interval-

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estimation time; it does not include the offline SHAP-based dimensionality reduction or model training time.

H. Performance Evaluation under PMU/WAMS Data Latency

To validate the effectiveness of the delay-aware training strategy for online SVSM prediction, simulations are conducted on the modified IEEE 39-bus system. For the modified Polish 2383-bus winter-peak system, the same workflow applies and the experiments are therefore not repeated. Assuming that the refresh interval of the proposed SVSM prediction method in the real-time security assessment (EMS) is 1 s, and considering that communication latency typically corresponds to 1–5 prediction refresh cycles, a communication delay of 5 s is assumed. Accordingly, the communication delay scenario is constructed by uniformly imposing a 5 s delay on all input features.

In the operational scenario of the modified IEEE 39-bus system, a 15-minute operating period is selected, generating a total of 3600 SVSM samples. These samples are partitioned into a training set of 2700 samples and a validation set of 900 samples, representing a 4:1 ratio, to test the proposed method. Three LightGBM training configurations are evaluated in this study: (1) a baseline model trained and tested using fully synchronized input–output pairs with no delay; (2) a delayed-input model that employs time-shifted inputs for prediction without applying the delay-aware training strategy; and (3) the proposed delay-aware model that incorporates both delayed inputs and the delay-aware training mechanism. A delay of 5 seconds is imposed by shifting the input features forward by five sampling intervals during training.

The experimental results are presented in Figs. 29 and 30. Among them, Fig. 29 illustrates the effect of applying a 5-s lag to the voltage magnitude at Bus 2, while Fig. 30 compares the prediction performance of the three training strategies.

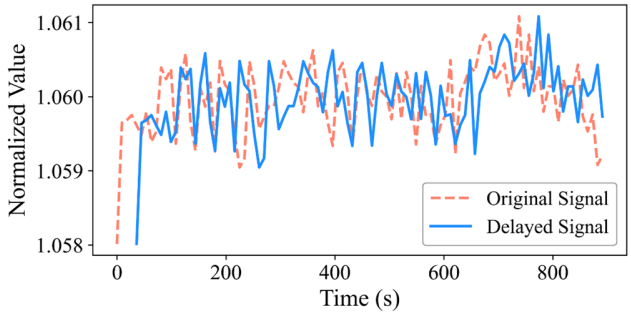


Fig. 29. Time-lag processing effect of voltage magnitude samples at Bus 2.

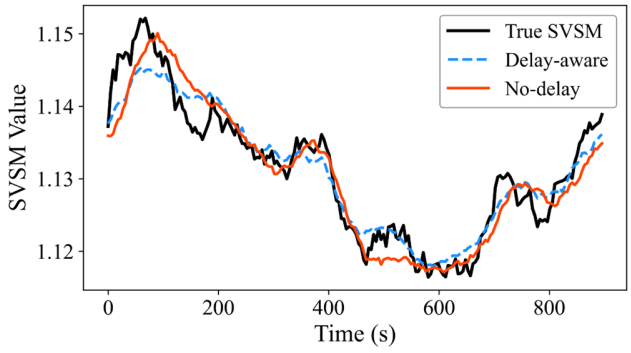


Fig. 30. Performance of models trained under three different strategies.

From Figs. 29–30, it can be observed that the PMU channel for measuring the voltage amplitude of Bus 2 has introduced a 5-second delay, and other input characteristics are the same. At this time, if the LightGBM model without considering the delay is still used for SVSM prediction, there is a significant deviation between the prediction result and the real value. After using the delay sensing training strategy proposed in this paper, the model can better capture the real change trend of SVSM and make the prediction results more consistent with the real curve.

I. Robustness Validation under System Topology Changes

To evaluate the effectiveness of the proposed voltage stability assessment scheme under different system conditions, various operating states were configured in the modified IEEE 39-bus system and the modified Polish 2383-bus winter-peak system, as summarized in Table VII. The table lists N–1 contingencies, N–2 contingencies, and complex fault scenarios for both systems. Following the same sample generation method used in the modified IEEE 39-bus system, three different validation sets were created, each containing 200 samples. The training dataset was assumed to remain unchanged, and the validation sets were used to test the prediction performance of the proposed algorithm under the new system states.

TABLE VII
PREDICTION OF DIFFERENT SYSTEM STATES

System	No.	System state	MAPE
Modified IEEE39-bus system	1	Disconnect lines 17-18	5.37%
	2	Disconnect lines 17-18 and lines 16-24	5.57%
	3	Disconnect lines 17-18, lines 16-24, and generator 30	5.75%
Modified Polish 2383-bus winter-peak system	1	Disconnect lines 1988-151	3.61%
	2	Disconnect lines 1988-151 and lines 2107-1628	3.84%
	3	Disconnect lines 1988-151, lines 2107-1628, and lines 1904-1681	4.02%

As shown in Table VII, tests were carried out on both the modified IEEE 39-bus system and the modified Polish 2383-bus winter-peak system under three different operating states. Although the prediction accuracy of the proposed algorithm decreases slightly in all cases, the reduction is within 5%, and the average accuracy remains above 95%. This demonstrates that the proposed method exhibits good robustness against system state variations. In addition, it can be observed that the large system possesses stronger redundancy: in the small system, the error increases from 2.8% to over 5%, whereas in the large system, it only rises from 3.3% to just above 4%. This indicates that when minor topological changes occur, the prediction performance of large-scale systems is less affected compared to that of small systems.

In addition, an incremental learning update strategy is designed in this study. This mechanism preserves the original tree structure and appends only a small number of new trees to the existing model and minor adjustments to selected weight parameters, allowing the parameters to incorporate information

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from newly encountered operating conditions and ensuring long-term model reliability.

The core idea of the proposed incremental learning strategy is as follows: during the offline stage, a comprehensive CPF simulation database is constructed to cover a wide range of representative operating conditions, including different renewable generation levels, load variations, generator switching events, and N–1 contingencies, providing the model with high-quality multi-scenario training data. During online operation, when the system detects a change in operating conditions, the model retrieves the most relevant samples from the offline database and performs fast incremental training, enabling the predictor to adapt to the new operating state within seconds. This approach eliminates the high computational latency associated with online CPF calculations while maintaining the model's capability for dynamic self-adaptation.

The proposed framework supports two modes for triggering model updates:

1) Routine Maintenance Mode:

Model updates are carried out periodically, with the update frequency determined by the variability of wind and solar generation, the stochastic nature of load behavior, and the overall system scale. Since wind speed and solar irradiance typically exhibit pronounced hour-level variations, and the loads considered in this study follow Gaussian fluctuations, routine update intervals of 1–3 hours are appropriate for small- to medium-sized systems. For large-scale systems, the interval can be extended to 3 hours or longer to reduce computational burden. Meanwhile, the system continuously performs continuation power flow (CPF) calculations in the background, storing the results in an offline database. To ensure the long-term reliability of model predictions, the system periodically compares the CPF results with SVSM forecasts (e.g., every 30 minutes). If the SVSM prediction accuracy fails to meet the required threshold (e.g., 95%) for several consecutive checks, an incremental model update is triggered.

2) Event-Driven Maintenance Mode:

When the system detects an abnormal event—such as generator switching, line outages, or other unexpected contingencies—an immediate model update is triggered. At this time, it is essential to ensure that the model can be updated rapidly while maintaining satisfactory prediction accuracy, so an appropriate update-window length must be selected to balance low computation latency with acceptable forecasting performance. In this study, the computation latency and window length were evaluated using the modified IEEE 39-bus system. For the modified Polish 2383-bus winter-peak system, the same workflow applies and the experiments are therefore not repeated.

For the IEEE 39-bus test case, a 20-minute operating period is selected, during which 3600 testing samples and 1200 validation samples are generated. The first 5 minutes are assumed to represent normal system operation, and an N–1 contingency corresponding to the disconnection of the 17–18 AC transmission line occurs at the 5-minute mark. To evaluate the effectiveness of the incremental learning strategy, datasets corresponding to 5-, 10-, and 15-minute post-contingency operating windows are retrieved from the CPF offline database, containing 150, 300, and 450 samples, respectively. The incremental learning computation latency and prediction accuracy are tested for each window length, and the simulation results are presented in Figs. 31–33.

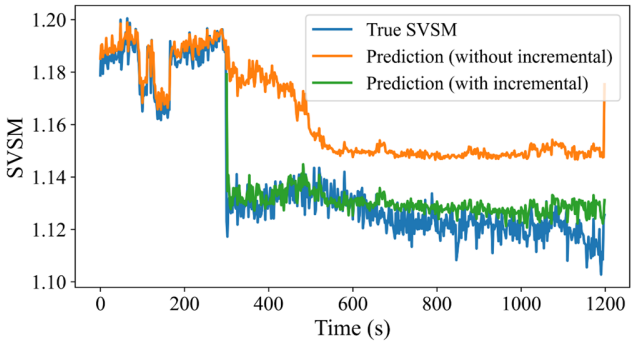


Fig. 31. Incremental learning effect under 5-minute time window.

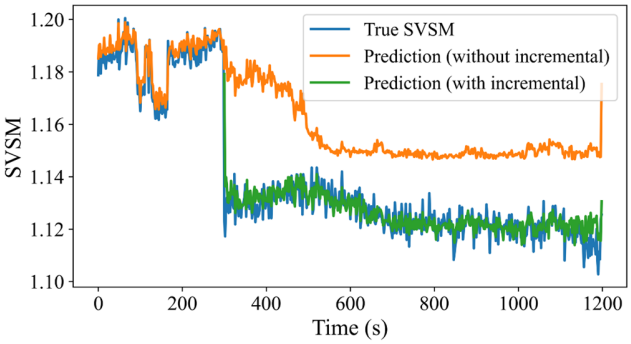


Fig. 32. Incremental learning effect under 10-minute time window.

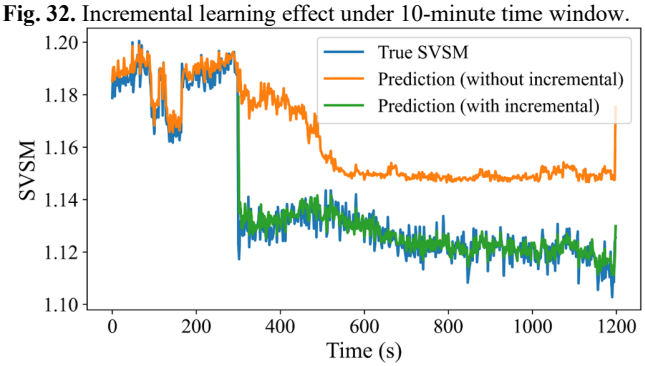


Fig. 33. Incremental learning effect under 15-minute time window.

From Figs. 31–33, it can be observed that when an N–1 contingency occurs at the 5-minute mark, the prediction accuracy of the model deteriorates significantly if the original training set is used without updating. In contrast, after performing rapid incremental training, the model quickly adapts to the post-contingency operating conditions, leading to a substantial improvement in SVSM prediction accuracy. The computation delays for the three window lengths are 0.04 s, 0.06 s, and 0.11 s, respectively, while the MAPE for predictions within the 5–20 minute range are 1.67%, 1.31%, and 1.27%. This indicates that while a larger update window yields higher prediction accuracy, it also results in greater computation latency. Therefore, among the three tested configurations, a 10-minute update window is recommended for rolling incremental training due to its optimal overall performance. The model should be refreshed every 10 minutes until the N–1 contingency

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is resolved, after which the system can revert to the routine maintenance mode with a reduced model update frequency.

H. Robustness Validation under Data Noise

Since noise in PMU measurements is unavoidable, the IEEE C37.118 standard requires that PMU measurement errors generally remain within 1%. Typically, a signal-to-noise ratio (SNR) of 20 dB corresponds to approximately 1% noise in the data. To evaluate the robustness of the proposed LightGBM–SHAP–KDE method under different noise environments, Gaussian white noise with an SNR ranging from 10 to 40 dB was added to the data, corresponding to a noise level of approximately 0.01%–3%. The test results are shown in Fig. 23 for both the modified IEEE 39-bus system and the modified Polish 2383-bus winter-peak system, where noise was introduced into both the training and testing datasets.

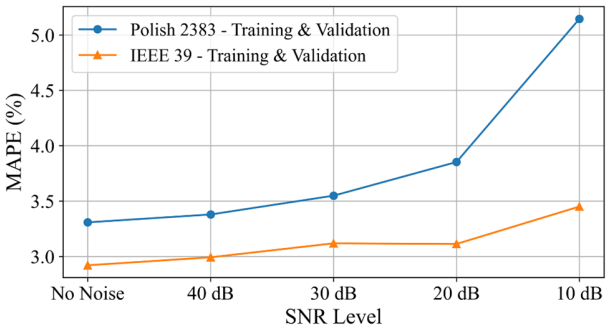


Fig. 34. MAPE of Predictions for Different Systems under Various Noise Scenarios.

From Fig. 34, it can be observed that the MAPE of the modified IEEE 39-bus system and the modified Polish 2383-bus winter-peak system are 2.8% and 3.3%, respectively, under noise-free conditions. Under the presence of 10 dB noise (corresponding to 3%), the maximum MAPEs of the two systems remain below 3.5% and 6%. These results indicate that the proposed algorithm possesses strong robustness and retains satisfactory predictive performance in noisy environments.

I. Robustness Validation under Missing Data

Due to the complexity of field environments, PMU measurement data are often subject to missing values caused by factors such as communication congestion, hardware failures, and natural or human interference. This poses higher requirements for the robustness of assessment models. In this paper, two types of data missingness are considered: missing completely at random (MCAR) and persistent feature loss. The former simulates occasional communication packet loss, whereas the latter represents complete measurement failures of certain features by specific PMU devices. In the experiments, 5%–30% of the data across all features in the training and validation sets were randomly removed, and 1–3 complete features were entirely omitted to test predictive performance. The results are illustrated in Fig. 35, which presents the prediction robustness of both the modified IEEE 39-bus system and the modified Polish 2383-bus winter-peak system under missing data conditions.

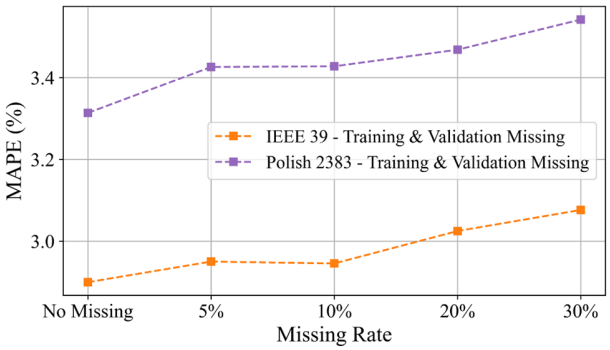


Fig. 35. MAPE of Predictions for Different Systems under Data Loss Scenarios.

From Fig. 35, it can be observed that in the modified IEEE 39-bus system and the modified Polish 2383-bus winter-peak system, when data missing rates ranging from 5% to 30% were tested, the prediction MAPEs of both systems fluctuated within 3%. This demonstrates that the proposed algorithm exhibits good robustness and maintains satisfactory prediction accuracy under missing data conditions. Moreover, in practical applications, interpolation techniques—such as linear interpolation, mean-value interpolation, and k-nearest neighbor (KNN) interpolation—are commonly employed to mitigate the impact of missing data. However, since the primary focus of this paper is to evaluate the robustness of the proposed algorithm, these methods are not discussed in detail here.

V. Conclusion

This paper proposed a data-driven framework for online SVSM estimation based on LightGBM–SHAP–KDE, designed to address the high-dimensional uncertainties introduced by renewable integration and VSC-HVDC. The approach enables efficient, accurate, and interpretable prediction of SVSM and its probabilistic distribution. In the modified IEEE 39-bus system, the proposed method achieved a mean absolute percentage error of 2.8% on 3,400 samples, with training and prediction times of 0.533 s and 0.017 s, respectively, demonstrating both high accuracy and computational efficiency. The probability distributions estimated by KDE closely matched the actual distributions, while SHAP analysis reduced the feature dimension from 144 to 55, improved interpretability, quantified the importance and directional influence of system variables, and provided simple control guidance for enhancing voltage stability. The scalability of the method is further verified using the modified Polish 2383-bus winter-peak system, together with a performance comparison against the traditional Cumulant interval-calculation method. Additional robustness tests, including data latency and asynchronism, network topology changes, measurement noise, and missing data, confirm that the proposed approach consistently maintains high prediction accuracy and fast computational performance under a variety of challenging operating conditions.

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